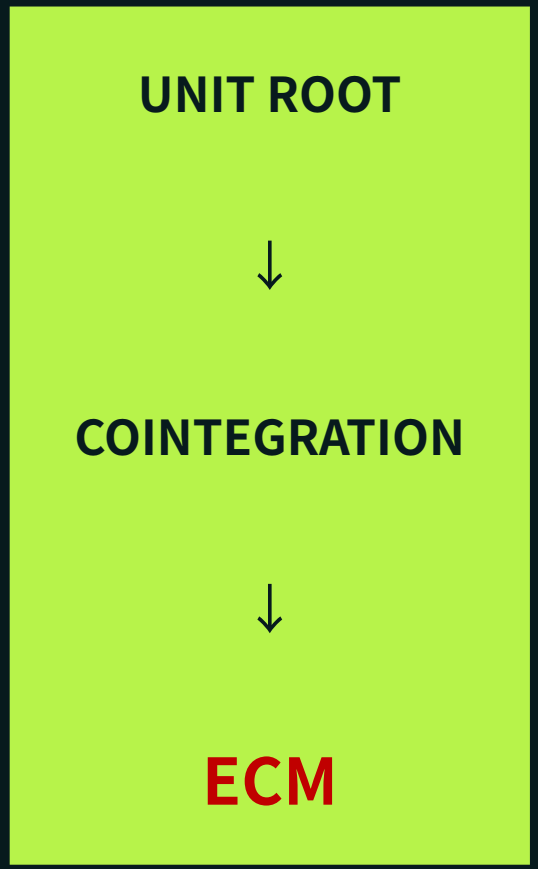
Cointegration & ECM Basics
Series that drift together

From unit roots and the ADF test to cointegration and the ECM



Advanced Financial Time Series · Week 7

Learn it in the order you handle non-stationary series; end with 5 assets

Spot the trend → test it → find series that wander together → forecast with that link

00	Unit roots and ADF	**Does this series wander?** Two trends · effects of non-stationarity · ADF
01	Cointegration	**The drunk and the dog** Definition · long-run equilibrium · example · stocks
02	Testing for cointegration	**Stationary residuals?** Engle-Granger · critical values · using Johansen
03	Error-correction model	**Drift apart, come back** ECM · consumption–income · pairs trading
04	5-asset example	**Find · test · signal** Screen · pair tests · Johansen · entry

Cointegration: the one exception to 'difference if non-stationary'

00

Unit Roots and the ADF Test

Unit Root and Augmented Dickey-Fuller Test — is this series stationary?

Trends come in two kinds: stochastic and deterministic

Trends: stochastic and deterministic — a set direction, or a wandering one?

Four standard models

$$
\begin{aligned}
y_t &= y_{t-1} + \varepsilon_t && \text{random walk} \\
y_t &= \alpha + y_{t-1} + \varepsilon_t && \text{random walk + drift} \\
y_t &= \alpha + \beta t + \varepsilon_t && \text{deterministic trend} \\
y_t &= \alpha + \beta t + \gamma\, y_{t-1} + \varepsilon_t, \quad |\gamma| < 1 && \text{deterministic trend}
\end{aligned}
$$

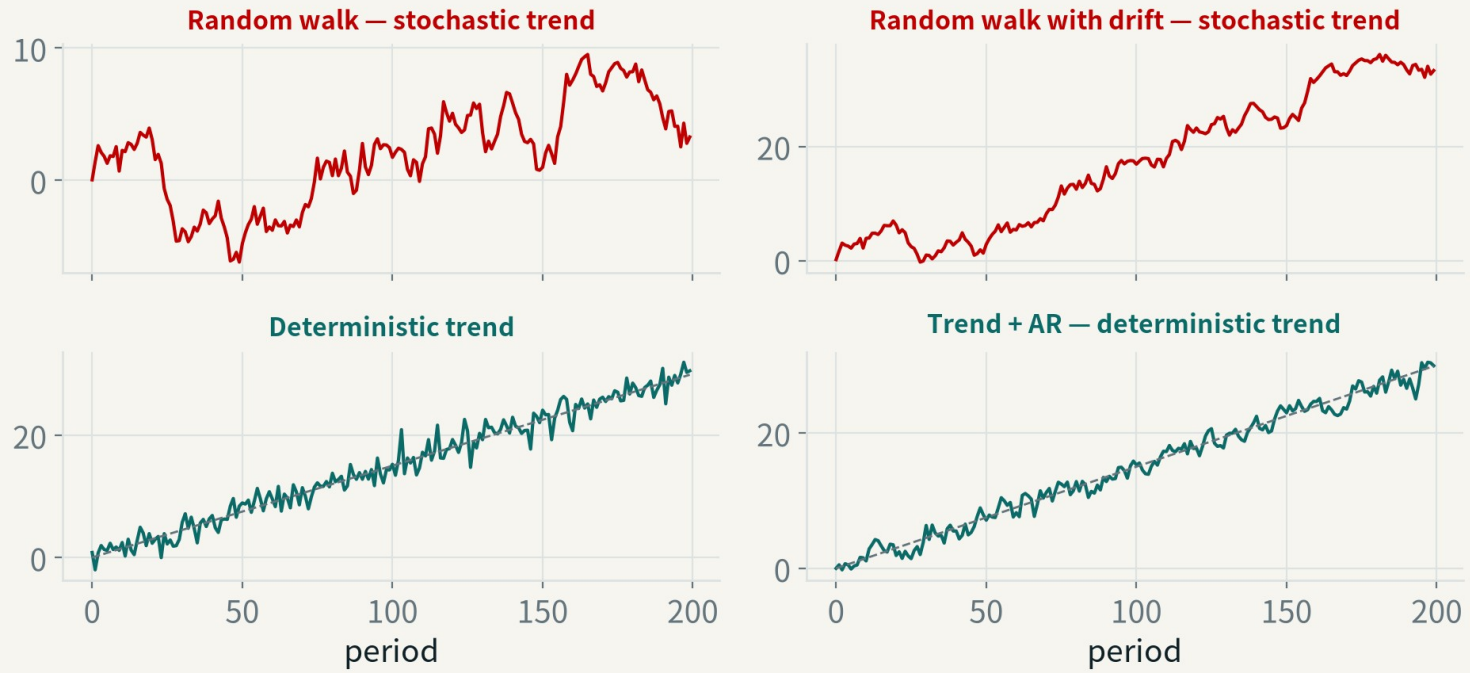
How to read it

- Top two: stochastic trend
- Shocks pile up and wander
- Bottom two: deterministic
- Swings around a line

Stochastic trend = unit root = random-walk type — this is the troublemaker

Stochastic trends wander; deterministic trends swing around a line

Four series from the same shocks — red: stochastic trend, green: deterministic (dashed: linear trend)



Hard to tell by eye — hence a test (ADF)

One difference removes a stochastic trend

Difference Δy(t) = y(t) − y(t−1) — look at changes, not levels

Differencing a drifting RW

$$y_t = \alpha + y_{t-1} + \varepsilon_t \;\Rightarrow\; \Delta y_t = y_t - y_{t-1} = \alpha + \varepsilon_t$$

How to read it

- Differenced: constant + noise → stationary
- Same idea as prices → returns
- Deterministic trend: regress on t

Drop the wandering levels; model the stationary changes

Regress non-stationary series and t- and F-tests cannot be trusted

Effects of non-stationarity — ordinary t and F tests hold only if all variables are stationary

Rewriting AR(1) in difference form

$$y_t = \alpha + \beta\, y_{t-1} + \varepsilon_t \qquad H_0 : \beta = 1$$
$$\Delta y_t = \alpha + (\beta - 1)\, y_{t-1} + \varepsilon_t$$
$$= \alpha + \rho\, y_{t-1} + \varepsilon_t, \qquad \rho = \beta - 1$$

Unit-root test (Dickey-Fuller)

$$H_0 : \rho = 0 \ \ (\text{unit root}) \qquad H_1 : \rho < 0$$
$$\text{reject if } t_{\hat{\rho}} < -2.9 \ \ (\text{not } -1.65)$$

How to read it

- β = 1 means a unit root (random walk)
- Test whether ρ = β − 1 = 0
- Critical value −2.9 — not the usual −1.65
- So test stationarity before estimating

Before any estimate, test 'is it stationary?'

In difference form, AR(2) also tests for a unit root with one ρ

Exercise — rewrite y(t) = α + β1 · y(t−1) + β2 · y(t−2) + ε(t) in Δy

Subtract y(t−1); add and subtract β2 · y(t−1)

$$\begin{aligned}\Delta y_t &= y_t - y_{t-1} = \alpha + (\beta_1 - 1)\, y_{t-1} + \beta_2\, y_{t-2} + \varepsilon_t \\ &= \alpha + (\beta_1 + \beta_2 - 1)\, y_{t-1} - \beta_2\, (y_{t-1} - y_{t-2}) + \varepsilon_t \\ &= \alpha + \underbrace{(\beta_1 + \beta_2 - 1)}_{\rho}\, y_{t-1} \underbrace{- \beta_2}_{\gamma}\, \Delta y_{t-1} + \varepsilon_t\end{aligned}$$

How to read it

- δ = α
- ρ = β1 + β2 − 1
- γ = −β2
- AR(p) the same way: p−1 lags of Δy

More lags add past Δy terms — the 'Augmented' DF

The ADF test uses one of two equations, with or without trend

Augmented Dickey-Fuller — add βt if the data trend clearly one way

No clear trend direction

$$\Delta y_t = \alpha + \rho\, y_{t-1} + \gamma_1 \Delta y_{t-1} + \cdots + \gamma_L \Delta y_{t-L} + \varepsilon_t$$

Clear trend direction

$$\Delta y_t = \alpha + \beta t + \rho\, y_{t-1} + \gamma_1 \Delta y_{t-1} + \cdots + \gamma_L \Delta y_{t-L} + \varepsilon_t$$

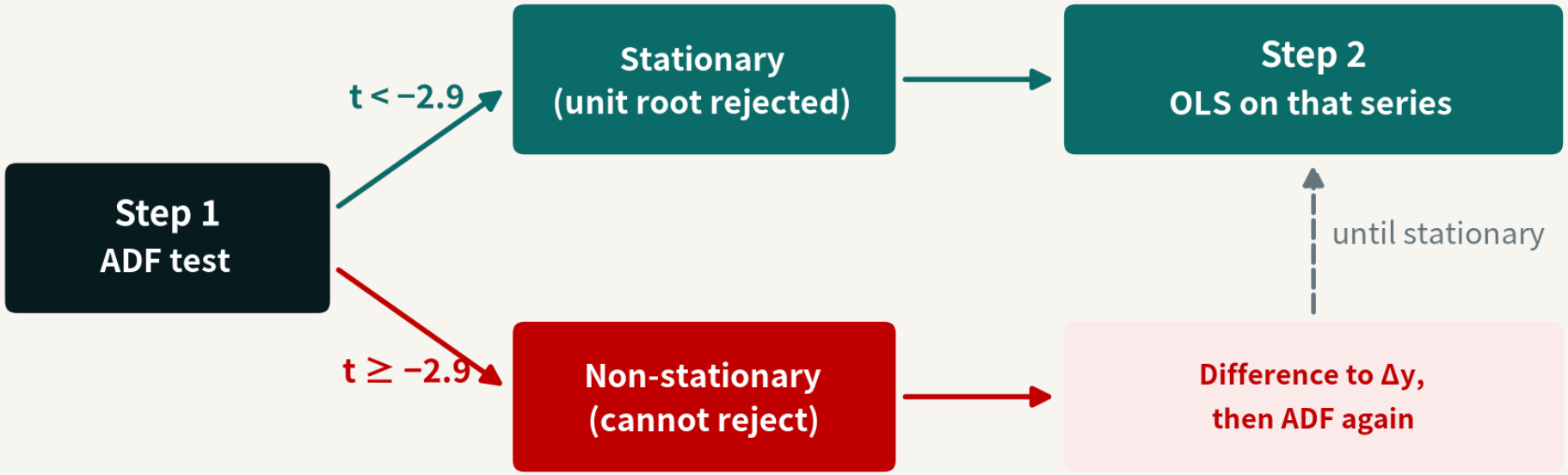
How to read it

- Null: ρ = 0 (non-stationary)
- t < critical value → reject → stationary
- No trend −2.9, with trend −3.5
- Lag L by AIC · BIC

ADF's null is 'non-stationary' — reject it for stationary

Depending on the test, use it as is or difference and re-test

Handling unit-root models — the same steps for y in an AR, for y and x in an ADL



The one exception: if x and y are cointegrated, do not difference

01

Cointegration

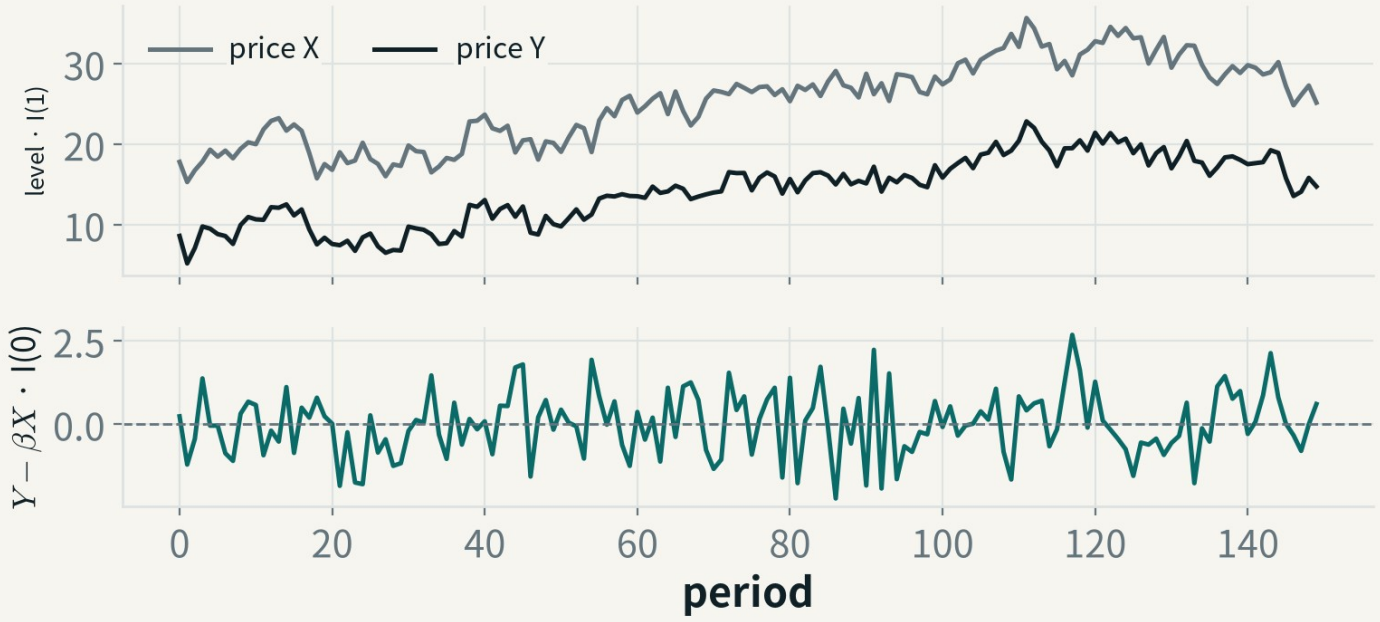
Cointegration — two series that each wander, but wander together

A drunk and a leashed dog each wander, but never drift apart

What is cointegration? — two drunks are unrelated, but a leash ties the distance

Analogy

- Person: random walk
- Dog: random walk
- Distance between: within a range
- → the positions cointegrate



Cointegration = each wanders, but the gap stays put

Cointegrated: a stationary linear mix of two non-stationary series

x and y each have a stochastic trend, but y − c·x is stationary — they share a common stochastic trend

Definition

$$x_t,\ y_t:\ \text{non-stationary}$$
$$y_t - c\,x_t:\ \text{stationary} \ \Rightarrow\ \text{cointegrated}$$

Long-run equilibrium

$$y_t = c\,x_t \qquad (\text{long-run equilibrium})$$

How to read it

- c: cointegrating coefficient (the tie ratio)
- y − c·x: deviation from equilibrium
- Deviations return to near 0
- Common stochastic trend

Cointegration = a long-run equilibrium of two series

If they share a hidden common trend z, z drops out of y − c·x

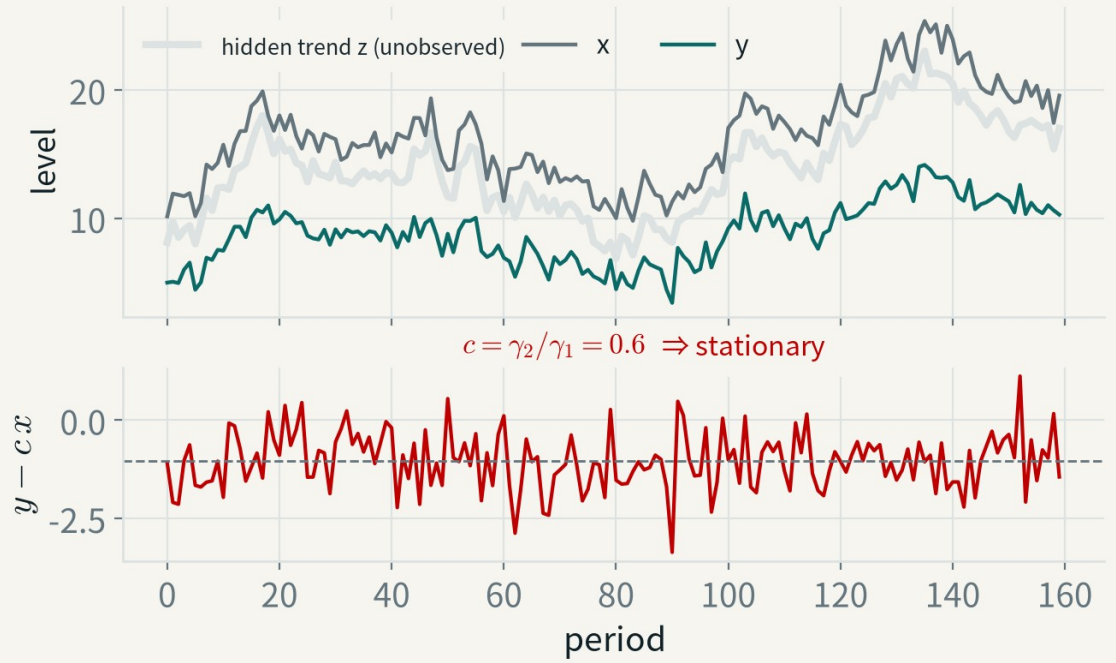
Test example — z is an unobserved random walk; x and y are observed. Show cointegration and find c

Setup

$$z_t = z_{t-1} + \varepsilon_{z,t} \qquad \text{(unobserved common trend)}$$
$$x_t = \alpha_1 + \gamma_1 z_t + \varepsilon_{x,t}$$
$$y_t = \alpha_2 + \gamma_2 z_t + \varepsilon_{y,t}$$

Answer

- Build the combination that cancels z
- What remains: constant + white noise



c is the ratio in which the two load on the common trend

Worked out, c is the ratio of the two loadings

Answer — find the combination that cancels z in both equations

Solution

$$\gamma_1 y_t - \gamma_2 x_t = (\gamma_1 \alpha_2 - \gamma_2 \alpha_1) + \underbrace{(\gamma_1 \varepsilon_{y,t} - \gamma_2 \varepsilon_{x,t})}_{\text{white noise}} \Rightarrow \text{stationary}$$

$$\gamma_1 y_t - \gamma_2 x_t = \gamma_1 \left(y_t - \frac{\gamma_2}{\gamma_1} x_t \right) \Rightarrow c = \gamma_2 / \gamma_1$$

How to read it

- Line 1: z drops out; stationary
- Line 2: factor out γ1 to read c
- E.g. γ1 = 1, γ2 = 0.6 → c = 0.6

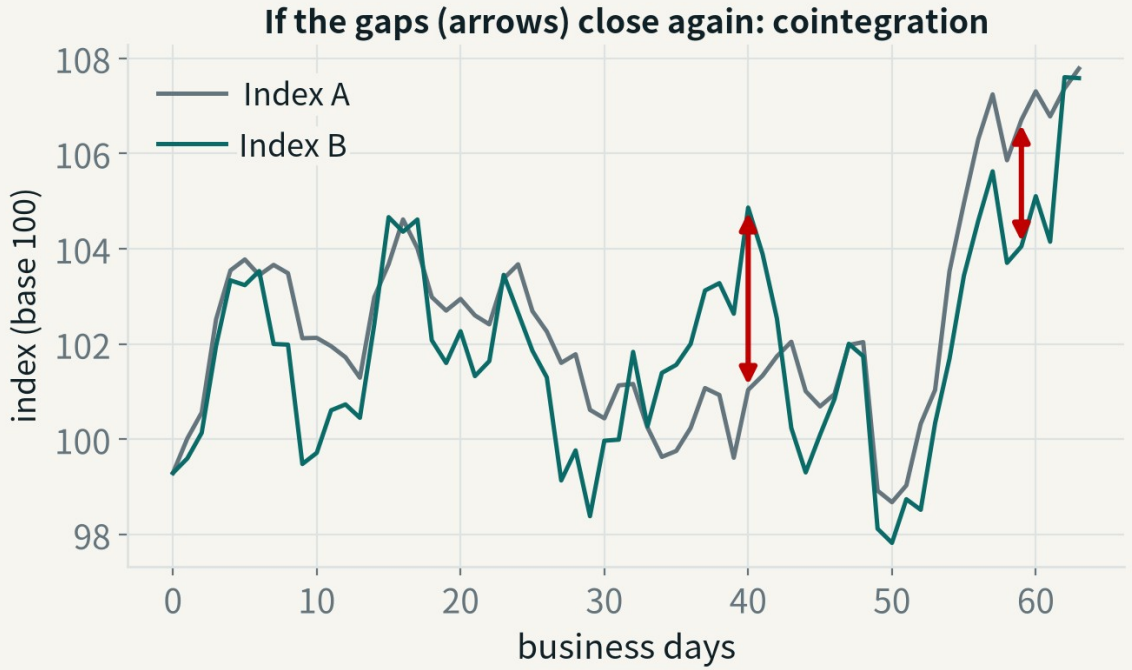
Ratio cancelling the trend = cointegrating vector

Cointegration underlies pairs trading in stocks

If two indices (e.g. KOSPI and Dow) are cointegrated — simulated example

How to read it

- Arrows: where the indices drift apart
- If cointegrated, it closes
- Sell the dear one + buy the cheap one
- A kind of arbitrage = pairs trading



But confirm cointegration with a test, not a chart

02

Testing for cointegration

Engle-Granger Test — are the regression residuals stationary?

Engle-Granger takes one regression and one ADF

Step 1: OLS on levels gives b and residuals e(t); step 2: ADF on them

Engle-Granger two steps

$$\text{Step 1:}\quad y_t = \alpha + \beta\, x_t + \varepsilon_t \xrightarrow{\text{OLS}} a,\ b, \qquad e_t = y_t - a - b\, x_t$$

$$\text{Step 2:}\quad \Delta e_t = \alpha + \rho\, e_{t-1} + \gamma_1 \Delta e_{t-1} + \cdots + \gamma_L \Delta e_{t-L} + \omega_t$$

$$t_{\hat{\rho}} < -3.4 \ \Rightarrow\ \text{cointegrated}$$

How to read it

- b: estimated coefficient (= c)
- e(t): estimated equilibrium error
- e stationary → cointegrated
- e non-stationary → spurious
- With trend: −3.8

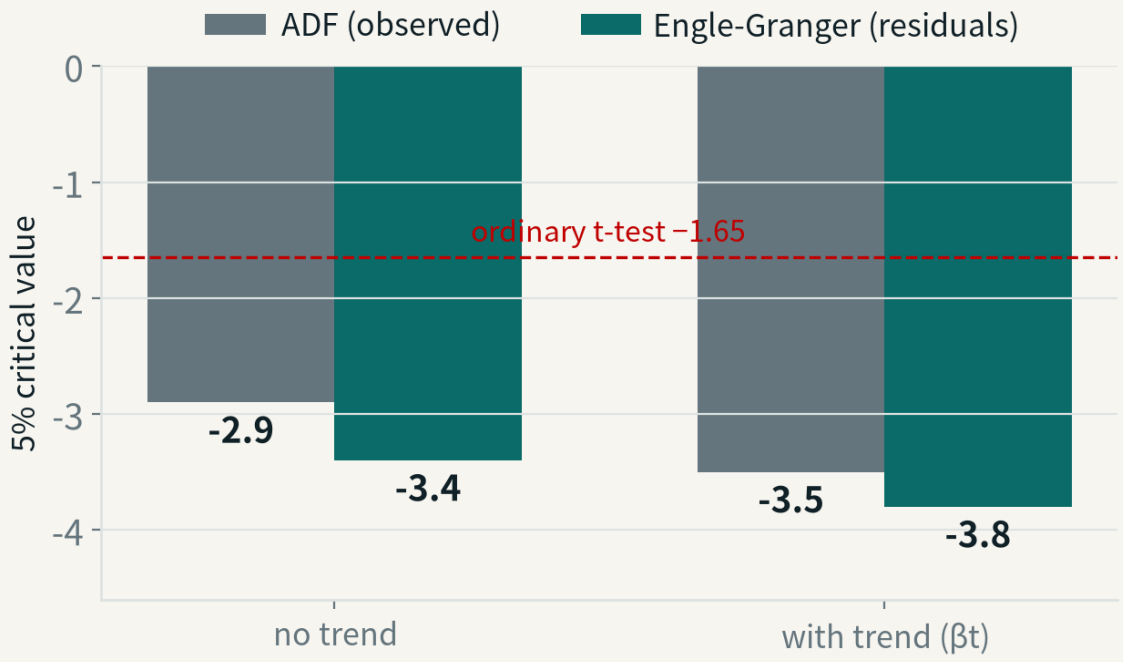
Cointegration test = residual unit-root test

Residuals need stricter critical values than ordinary ADF

OLS picks b to shrink the residuals, so they tend to look stationary

5% critical values

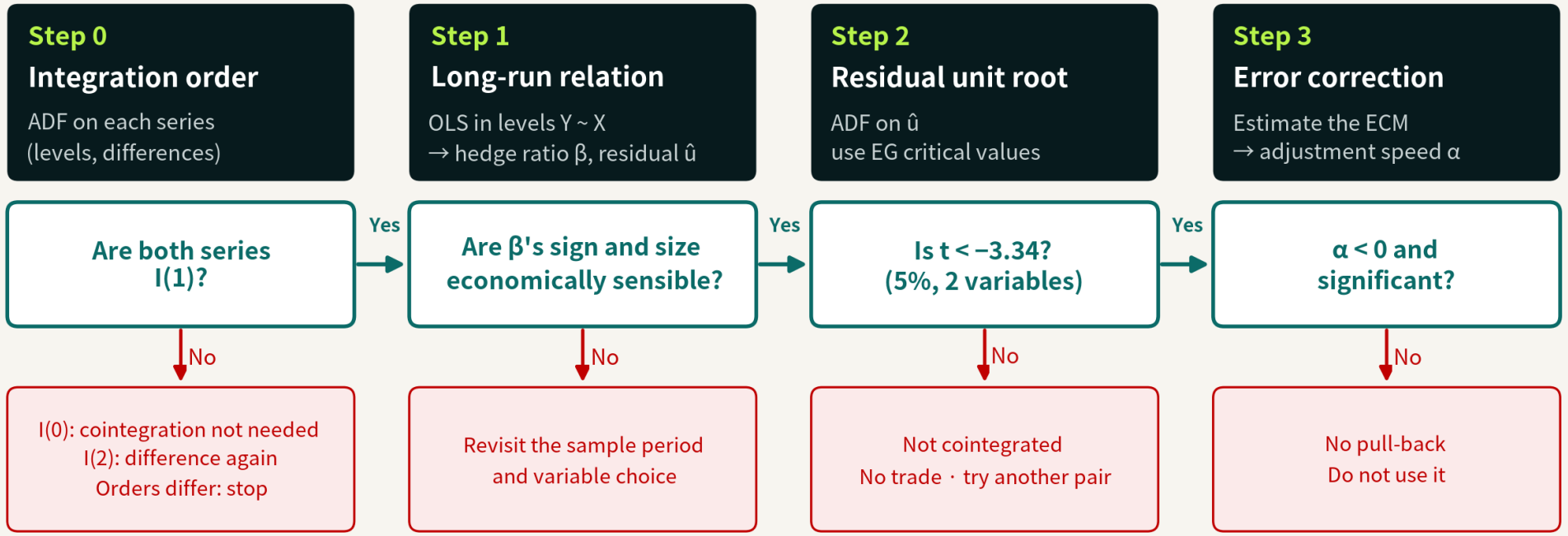
- ADF −2.9 → EG −3.4
- With trend −3.5 → −3.8
- Between −2.9 and −3.4: not cointegrated



Using −2.9 on residuals finds false cointegration

From step 0 to 3, move on only on a 'yes'

ADF steps above (step 0) + Engle-Granger two steps + error-correction check (step 3)



Use the relation for forecasting and trading only after four 'yes'es

Three cointegration tests are standard

Methods of Testing for Cointegration

ENGLE-GRANGER

Regression + ADF

- Simplest, for two variables
- Limited with 3+ variables

JOHANSEN

Many variables at once

- Also counts relations
- Needs a large sample

PHILLIPS-OULIARIS

Refines EG

- Residuals do not follow DF
- Uses the PO distribution

Two variables: Engle-Granger; many: Johansen

Johansen takes many assets at once and tells you how many relations there are

Engle-Granger goes pair by pair, Johansen takes k at once — leave theory (VECM) to software; learn inputs and outputs

INPUT

Inputs

- k prices, each I(1)
- Number of lags (usually 1–2)
- Constant or not

OUTPUT

Outputs

- Trace stats, critical values
- Cointegrating vectors (weights)

READ

What to read

- Number of relations = rank r
- $r \geq 1$: r spreads

Johansen answers two things: 'how many' (rank) and 'what ratio' (vector)

One line of code, three outputs to read

Python statsmodels — prices is a price table, one column per asset (I(1) checked in step 0)

PYTHON

```
from statsmodels.tsa.vector_ar.vecm import coint_johansen

res = coint_johansen(prices, det_order=0, k_ar_diff=1)

res.lr1         # ① trace statistics
res.cvt[:, 1]   # ② 5% critical values
res.evec[:, 0]  # ③ first cointegrating vector
```

How to read it

- det_order=0: with constant (default for prices)
- k_ar_diff: VAR lags − 1
- Rows with ① > ②: reject; stop at the first ① < ②
- ③ ÷ its first element: hedge ratios

Compare ① ② from the top to set the rank; build the spread from ③

Read the trace test up from r = 0; stop at the first non-rejection

E.g. 4 assets (k = 4), T = 250 days — software output as a table

Null	trace ①	5% ②	Decision
r = 0	63.2	47.85	**Reject → go on**
r ≤ 1	22.5	29.80	**Stop = rank 1**
r ≤ 2	7.1	15.49	—
r ≤ 3	2.0	3.84	—

※ Rank 0: no relation · 1–3: number of spreads · 4 (= k): recheck step 0

Trace statistic (for reference: how it is computed)

$$\text{trace}(r) = -T \sum_{i=r+1}^{k} \ln\left(1 - \hat{\lambda}_i\right)$$

$\hat{\lambda}_i$: eigenvalues from software

The r in the first row with '① < ②' is the number of relations

03

Error-correction model

Error Correction Model — a gap is closed in the next period

If cointegrated, estimate an ECM instead of differencing

Explain Δy(t) by yesterday's deviation from equilibrium plus short-run changes

ECM

$$\Delta y_t = \alpha + \underbrace{\beta_1 \left(y_{t-1} - b\, x_{t-1}\right)}_{\text{error correction, } \beta_1<0} + \beta_2 \Delta y_{t-1} + \beta_3 \Delta x_{t-1} + \varepsilon_t$$

How to read it

- (y − b·x): yesterday's gap
- β1 < 0: pulls back by the gap
- β2, β3: short-run effects
- More lags allowed

Differencing alone drops the long run — the ECM keeps both

Consumption and income: temporary shocks fade; permanent ones move to a new equilibrium

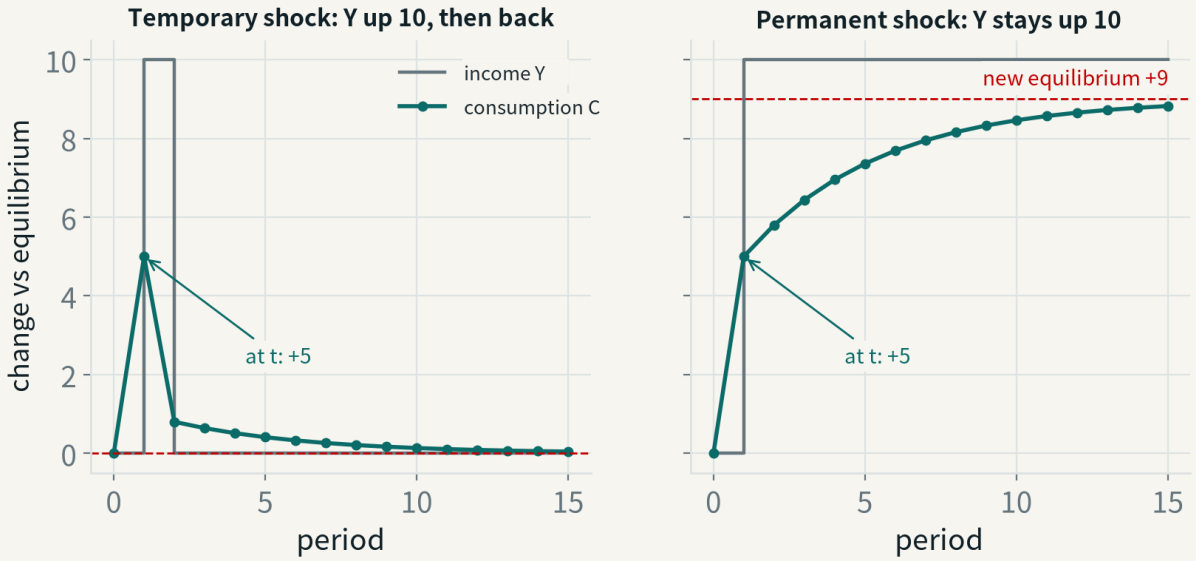
Long-run APC 0.9, short-run MPC 0.5, gap to equilibrium shrinks 20% each period

Long run and the ECM

$$\text{long run:} \quad C_t = 0.9\, Y_t$$
$$\text{ECM:} \quad \Delta C_t = 0.5\, \Delta Y_t - 0.2\,(C_{t-1} - 0.9\, Y_{t-1}) + \varepsilon_t$$

How to read it

- Consumption +5 at the shock, then to 0 or +9



The ECM puts 'short-run reaction' and 'return to equilibrium' in one equation

Shrinking 20% a period, half the gap closes in about 3 periods

Half-life — the error-correction coefficient as 'how many days to come back'

Half-life

$$\text{gap}_t = 0.8 \times \text{gap}_{t-1}$$

$$h_{1/2} = \frac{\ln 0.5}{\ln 0.8} \approx 3.1 \text{ periods}$$

How to read it

- −0.2: fixes 20% of the gap each period
- Remaining gap × 0.8 each time
- Shorter half-life, faster return
- Sets the pairs-trading holding period

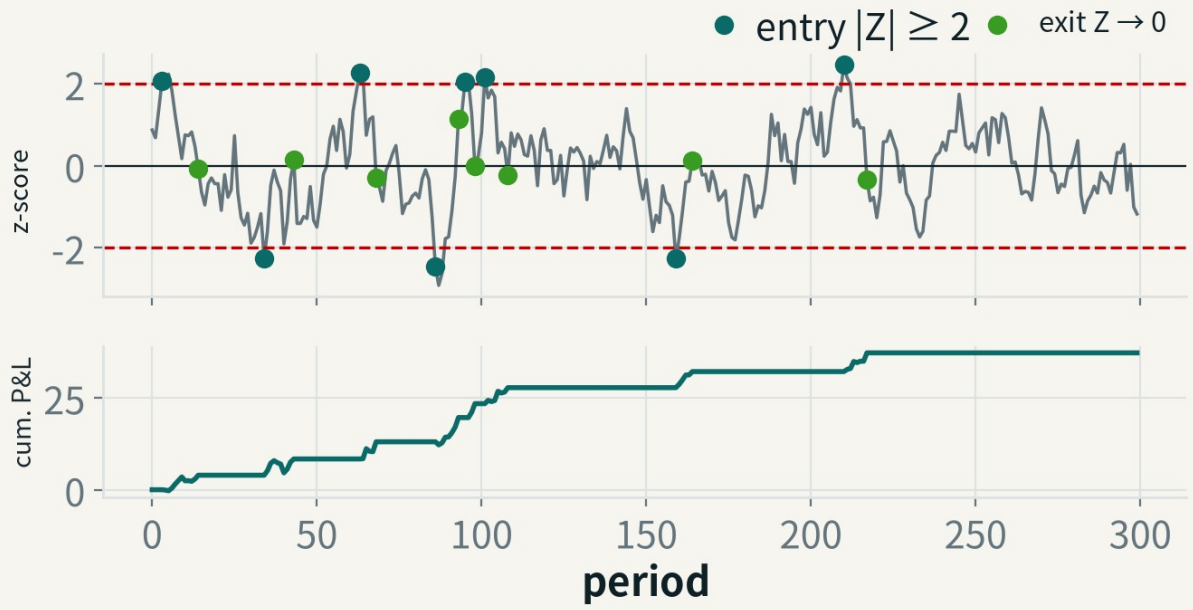
The more negative the coefficient, the faster it returns

Pairs trading enters when the gap is wide and exits at 0

Enter when the spread z-score passes ±2, exit near 0 — simulation

How to read it

- Top: spread z-score
- Bottom: cumulative P&L
- Profits while cointegration holds



You trade the 'gap', not the prices

When cointegration breaks, the gap does not come back

Structural breaks — mergers, regulation, business shifts (simulation: breaks after t = 180)

Caution

- Past cointegration is no guarantee
- Stop-losses and re-testing



Pairs trading without a stop-loss is a bet on a random walk

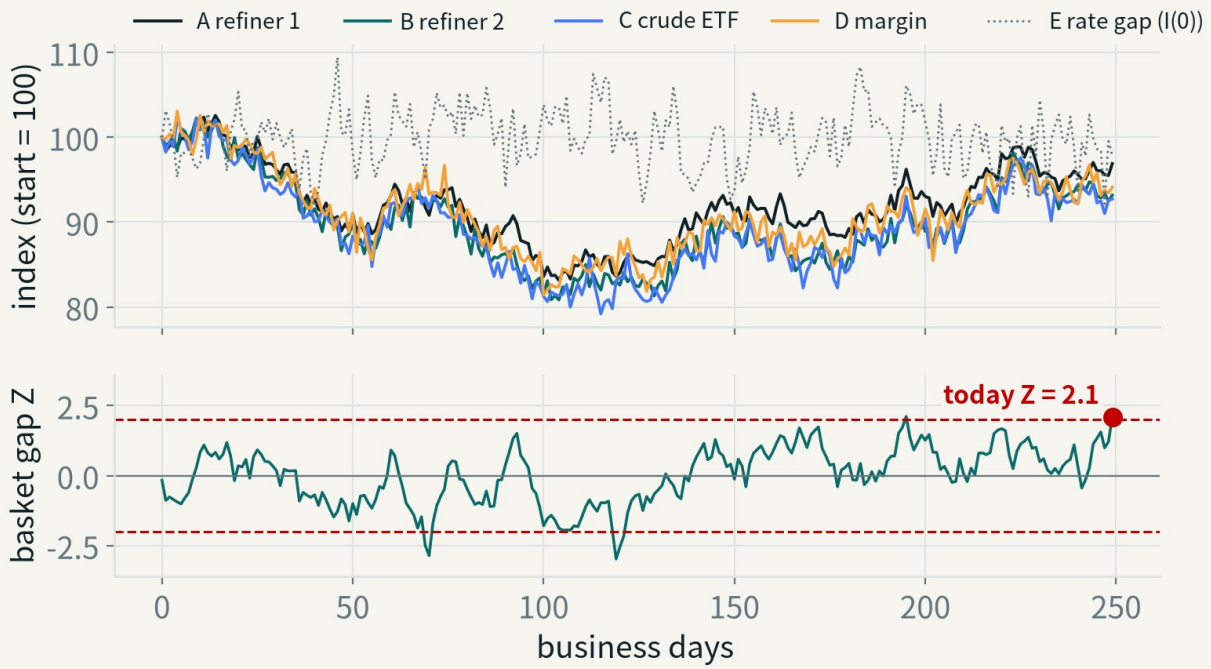
04

A 5-Asset Pairs-Trading Example

Screen → pair tests → system test → today's signal (hypothetical)

Of five assets, four wander together and one is already stationary

Hypothetical — A refiner 1, B refiner 2, C crude-oil ETF, D crack-spread index, E rate spread · 250 days



Goal: find a tradable 'gap' and decide whether to enter today

Step 0: E is already stationary, so drop it; keep A–D

ADF t-stats, with constant — below the 5% value −2.9 rejects the 'non-stationary' null

Asset	Level ADF	Diff. ADF	Decision
A	−1.42	−15.3	**I(1)**
B	−1.87	−14.8	**I(1)**
C	−2.10	−16.1	**I(1)**
D	−0.95	−13.9	**I(1)**
E	−4.12	—	**I(0), dropped**

How to read it

- A–D: levels not rejected, differences rejected → I(1)
- E: rejected in levels → stationary
- Stationary E is no cointegration candidate

Five in, four go on to the next step

Engle-Granger: of six pairs, only A–B and A–D are cointegrated

ADF on each pair's residuals — EG 5% critical value −3.4

Pair	Residual ADF	ADF cut −2.9	EG cut −3.4
A–B	−3.71	Pass	**Cointegration**
A–C	−2.95	Pass	**No**
A–D	−3.52	Pass	**Cointegration**
B–C	−2.40	Out	**No**
B–D	−3.05	Pass	**No**
C–D	−1.80	Out	**No**

How to read it

- −3.71, −3.52 < −3.4 → cointegrated
- Two red rows: −2.9 gives false cointegration
- Six tests → beware lucky passes

Ordinary ADF values on residuals wrongly pick A–C and B–D

A–B: half-life 6.6 days; today's gap is near 0, so no signal

Step-1 OLS: intercept 10, hedge ratio 0.95 · today A = 100, B = 95 · ECM speed −0.10

Gap and half-life

$$e_t = A_t - 10 - 0.95\,B_t$$

$$\text{today}: \quad e = 100 - 10 - 0.95 \times 95 = -0.25$$

$$h_{1/2} = \frac{\ln 0.5}{\ln 0.90} \approx 6.6 \text{ days}$$

How to read it

- Gap −0.25 ≈ 0
- → wait today
- If it widens, half returns in a week

Pair found, but enter only on days the gap is wide

Johansen: with A–D together, there is one relation

k = 4, 2 lags (k_ar_diff = 1), with constant — as in the usage above

Null	trace	5%	Decision
r = 0	63.2	47.85	**Reject**
r ≤ 1	22.5	29.80	**Stop → rank 1**

※ First cointegrating vector normalised on A: (1, −0.62, −0.25, −0.10)

How to read it

- One long-run equilibrium ties the four
- Weights: A 1, B 0.62, C 0.25, D 0.10
- A basket spread wider than the A–B pair

Johansen sees the four as one bundle, not pairs

The basket gap has Z = 2.1, so today is an entry signal

Today A = 100, B = 95, C = 60, D = 40 · past basket gap mean 20.0, s.d. 1.0

Gap → Z-score → signal

$$z_t = A_t - 0.62\,B_t - 0.25\,C_t - 0.10\,D_t$$

$$\text{today}: \quad z = 100 - 58.9 - 15.0 - 4.0 = 22.1$$

$$Z = \frac{22.1 - 20.0}{1.0} = 2.1 \;\geq\; 2 \;\Rightarrow\; \text{enter}$$

How to read it

- Sell 1 unit of A
- Buy B 0.62 · C 0.25 · D 0.10
- Exit when Z is near 0
- Z above 4: stop-loss and re-test

All figures hypothetical — in practice, regular re-testing and stop-losses go with it

05

Synthesis

Test → difference, or → error-correct

Non-stationary series are handled by whether they cointegrate

Start with ADF, then branch on the result

Situation	To do	Model
Stationary (ADF rejects)	Use as is	AR · ADL · VAR (levels)
Non-stationary, not cointegrated	Difference it	AR · ADL · VAR (differences)
Non-stationary, cointegrated	**Diffs + error term**	**ECM · VECM**
Cointegration breaks	Drop the relation	Stop-loss · re-test

Next week: state-space models, Kalman filter

Does it wander? — ADF

Do they wander together? — Engle-Granger

Does it come back? — ECM

Next week
State-space and Kalman filter